Cella: a serverless web resource for querying discriminative cell-type identity across twelve human organ atlases

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Abstract

Supervised cell-type classifiers trained on curated single-cell atlases learn, for every cell type, a weight for every gene, yet these weights are distributed only inside model files. We present Cella, an open web resource that turns the logistic-regression classifiers and pseudobulk expression of the CellTypist organ atlases (12 organs, 38 datasets, 3.69 million cells, 399 organ-specific labels harmonised to 145 Cell Ontology terms) into five interactive tools: an organ browser, a discriminative marker atlas, a cross-organ fingerprint explorer, a cell-type relationship graph and a gene-signature lookup. Building the resource exposed a pitfall in reading classifier coefficients directly: because genes are standardised before fitting, genes detected in almost no cells acquire large weights for rare classes, and 58% of raw top-25 positive coefficients were genes expressed in fewer than 10% of the cell type's own cells. Gating markers on expression prevalence restored canonical markers and, on 48 leave-one-dataset-out splits, produced marker panels better than raw coefficients and not significantly different from Wilcoxon differential expression in a classifier-free readout. We benchmarked the signature lookup against 542 external marker sets, with one library used for development and three held out; ranking by the fraction of cells expressing the query placed the correct cell type first for 40% of held-out queries, against 17% for mean classifier weight, and expression-based scoring beat mean weight in every library, so classifier weights are used to explain hits rather than to rank them. A calibrated confidence call and an ontology-aware treatment of related cell types accompany every lookup. Cella runs entirely in the browser from precomputed files, so it is hosted at no cost and can be archived with its code; the same codebase also serves a FastAPI backend. Cella is available at `https://mdjubayerhossain.com/cella/` and the source at `https://github.com/hossainlab/cella`.

Keywords: single-cell RNA sequencing; cell-type annotation; marker genes; CellTypist; Cell Ontology; web server

1 Introduction

Large single-cell reference atlases have shifted cell-type annotation from manual inspection of clusters towards automated label transfer [1, 2]. Among the most widely used tools is CellTypist, which fits regularised logistic-regression classifiers to harmonised reference data and was first applied to immune cells across sixteen human tissues [3]. Its successor work harmonised cell-type labels across Human Cell Atlas datasets and released organ-level atlases with matching classifiers [4, 5]. Each of these classifiers is, in effect, a matrix with one weight per gene and cell type, describing which genes the model uses to separate that cell type from its neighbours in the same organ.

Atlas portals such as CZ CELLxGENE Discover [6] and reference-mapping services such as Azimuth [7] make expression and annotation accessible at scale, and curated databases such as CellMarker [8] and PanglaoDB [9] catalogue literature markers. The classifier weights themselves, however, are distributed inside serialised model files. Reading them requires programming, and comparing them across organs requires harmonising labels, aligning gene spaces and handling cell types that are annotated at different granularities

in different organs. Questions that the weights could answer directly—which genes define a cell type in a given organ, which of those are shared across organs, and which cell type a list of genes points to—are therefore not easily asked by most users.

We built Cella to make these questions answerable in a browser. In doing so we found that classifier coefficients cannot be read naively as marker rankings, and that a resource built on them needs to be benchmarked rather than assumed to work. This article describes the resource, the prevalence gating that makes its marker lists usable, two benchmarks on held-out data and external marker sets that determined how its tools score, a cross-organ use case, and a serverless deployment that keeps the resource free to host and simple to archive.

2 Materials and methods

2.1 Data sources and harmonisation

We used the twelve adult human organ atlases distributed by CellTypist (blood, bone marrow, heart, hippocampus, intestine, kidney, liver, lung, lymph node, pancreas, skeletal muscle and spleen) and their pretrained classifiers [4, 5]. Each atlas provides cell-level metadata, including a curated annotation on which the classifier was trained and a Cell Ontology label; each classifier provides a coefficient matrix over its own feature set (5,529–10,396 genes) together with the standard scaler fitted to its training data. Model downloads are scripted and recorded with SHA-256 checksums. Each curated label was assigned the most frequent Cell Ontology label among its cells, which was mapped to a Cell Ontology identifier [10] by exact label, then synonym, then a single manual override, which resolved all 145 terms used across the 399 organ-specific labels; 49 terms occur in two or more organs. For every organ we computed a pseudobulk summary per curated label, namely the mean log-normalised expression and the fraction of cells with non-zero expression, streaming the expression matrix in blocks so that no atlas is loaded into memory whole.

2.2 Prevalence-gated discriminative markers

For a curated label c in organ o, let $\beta_{c,g}$ be the classifier coefficient of gene g, $p_{c,g}$ the fraction of cells of c expressing g, and $\bar{p}_{-c,g} = \max_{c' \neq c} p_{c',g}$ the highest such fraction among the other labels of the organ. Positive markers are the top n genes by $\beta_{c,g}$ among those with $p_{c,g} \geq \tau$; negative markers are the bottom n genes by $\beta_{c,g}$ among those with $\bar{p}_{-c,g} \geq \tau$, because a negative weight is informative only if some other cell type in the organ expresses the gene. We stored $n = 25$ markers per label with $\tau = 0.1$; setting $\tau = 0$ recovers the raw coefficient ranking. For each Cell Ontology term, the marker atlas takes the top ten gated positive markers in each organ where the term occurs and reports genes present in every such organ (universal core), genes shared by two or more organs, and genes found in a single organ (tissue-adapted).

2.3 Relationship graph and organ-level layers

The relationship graph links Cell Ontology terms whose classifier-weight vectors are similar. A term's vector is the unweighted mean of the weight rows of all curated labels annotated with it, over the union of genes; each gene is averaged over the rows whose model contains it, and genes absent from every row are set to zero. Each term proposes its six most similar terms with cosine similarity of at least 0.15, an edge is kept if either end proposes it (so a node can have more than six neighbours), and node positions come from a seeded spring layout so that the graph is identical on every load. The organ browser adds the atlas UMAP embedding (downsampled to 25,000 cells), transcription-factor activities inferred with the univariate linear model of decoupleR [11] over CollecTRI regulons [12], gene-set activities for six MSigDB collections (release 2023.2.Hs, including the hallmark collection [13]), and ligand–receptor scores computed as the product of mean ligand and receptor expression between labels, using the consensus resource of LIANA [14] and requiring each partner to be expressed in at least 10% of cells.

2.4 Gene-signature lookup

Given a query gene set Q, Cella scores every curated label c in every organ o by

$$S(c) = \frac{1}{|Q_o|} \sum_{g \in Q_o} p_{c,g}, \tag{1}$$

where Q_o is the part of the query present in organ o's expression vocabulary, and ranks labels across all organs. Because the score is averaged over Q_o rather than Q, an organ that matches fewer query genes is not penalised for the missing ones. Each hit reports, for its five most widely expressed query genes, both $p_{c,g}$ and the classifier weight $\beta_{c,g}$. Let S_1 be the best score and S_2 the best score of any cell type that is neither the same term nor a close ontology relative of it ($S_2 = 0$ if there is none). A lookup is called confident when $S_1 \geq 0.3$, $(S_1 - S_2)/S_1 \geq 0.1$, and the top hit matches at least two query genes (or all of them, for shorter queries). Close relatives are specific parents (ancestors of at most three atlas terms) and, for terms with at most five atlas descendants, their descendants; generic terms such as *cell* therefore have no close relatives. How the scoring rule and thresholds relate to the benchmark below is stated there.

2.5 Benchmark of signature lookup against external marker sets

Queries were cell-type marker sets curated independently of CellTypist, obtained from four libraries distributed by Enrichr [15]: Tabula Sapiens [16], Azimuth 2023, CellMarker 2024 and PanglaoDB Augmented 2021 (3,764 terms in total). We kept terms from tissues covered by the atlases (restricted to human entries for CellMarker; the PanglaoDB library does not label species), resolved their cell-type names to Cell Ontology identifiers (829 terms), and scored a query only if an atlas term could answer it and the query term was specific enough to be informative (at most five atlas descendants), leaving 542 queries. Queries whose source data overlap the CellTypist training data (Tabula Sapiens bone marrow, muscle and spleen; Azimuth lung, heart and kidney) were analysed separately (161 queries), leaving 381 independent queries.

Each method ranks Cell Ontology terms by the best score of any curated label annotated with the term, so the organ is not given. A hit is correct if it is the query term, one of its descendants, or a specific ancestor. The rank of a query is one plus the number of incorrect terms scoring above the best correct term, plus one half for each incorrect term tied with it, so that methods producing many ties are not favoured. We compared nine scoring rules: mean classifier weight; mean weight after prevalence gating; per-gene z-scored weights with and without gating; the classifier's softmax probability for a pseudo-cell expressing only the query genes; mean fraction expressing; mean per-gene z-scored expression; a hypergeometric test against the gated top-25 marker lists; and the average of the z-scored per-gene-standardised gated weight and fraction-expressing scores. CellMarker (150 independent queries) served as the development library and the other three libraries (231 queries) were held out. Scoring was restricted to genes in the classifiers' feature sets, so that weight- and expression-based rules see the same genes; the deployed lookup uses the full expression vocabulary. We report top-1 and top-5 accuracy with 95% percentile bootstrap intervals (2,000 resamples over queries), mean reciprocal rank (MRR), and exact McNemar tests on paired top-1 outcomes. To mimic short pasted lists, each query was also subsampled to 3, 5 and 10 genes in five random draws (queries with fewer genes than the subset size were skipped).

2.6 Benchmark of marker panels on held-out datasets

To ask whether gated coefficients make better marker panels than differential expression, we held out each source dataset of each organ in turn (48 splits over 12 organs). Training cells came from the remaining datasets, capped at 500 per label, for all labels with at least 50 training cells; test cells came from the held-out dataset, capped at 300 per label, for labels with at least 20 test cells, and splits with fewer than three testable labels were skipped. Sampling used a single fixed seed. Within each split, and on training cells only, we retrained CellTypist (no feature selection) on the released model's feature set and ranked markers by raw coefficient, by gated coefficient, by Wilcoxon rank-sum score in Scanpy [17] (one label versus the

rest, restricted to the same feature set) and by gated log fold change; prevalence for gating was computed on training cells. For $k \in \{3, 5, 10, 20\}$ markers per label we measured macro-F1 on the held-out cells (over the union of true and predicted labels) in two ways: a classifier-free assignment of each cell to the label whose own k markers have the highest mean z-score (standardised with training-cell means and deviations), and a fresh multinomial logistic regression [18] trained on the union of all labels' markers. Differences between methods were tested with paired Wilcoxon signed-rank tests across splits, and P values were Holm-adjusted over all 24 comparisons (two readouts, four panel sizes, three comparators); we report adjusted values (P_{adj}). Splits from the same organ share most training cells, so they are not fully independent.

2.7 Cross-organ invariance of discriminative identity

For each Cell Ontology term annotated in at least two organs, we compared its weight vectors between every pair of organs by cosine similarity over the genes shared by the two models (labels sharing a term within an organ were averaged, weighted by cell count). The null for an organ pair is the distribution of cosines between the term-level vectors of different Cell Ontology terms in the same two organs, and the calibrated invariance of a term is the mean, over its organ pairs, of the cosine minus that pair's null mean. Terms that are ontology ancestors of another annotated term were scored only over the organs in which no descendant was annotated separately, or excluded when fewer than two such organs remained. Terms were grouped by their first matching Cell Ontology ancestor into haematopoietic lineages (lymphoid, myeloid, erythroid/megakaryocytic) and structural lineages (epithelial, endothelial, mesenchymal/stromal); other lineages were not tested. Groups were compared with two-sided Mann–Whitney tests. A peripheral variant used only organ pairs in which neither organ is blood, bone marrow, spleen or lymph node. Coefficient reproducibility was estimated by retraining each organ's classifier on five disjoint, equally sized folds of 200 cells per label and computing the mean pairwise cosine of the replicate vectors; labels with fewer than 1,000 cells could not be split this way and were not scored.

2.8 Implementation and deployment

The offline pipeline is written in Python and writes Parquet artifacts; a FastAPI service exposes them, and a Next.js application provides the interface. The interface can be built in two modes from the same code. In server mode it queries the FastAPI service. In static mode, used for the public site, a command (`cella export`) calls every endpoint whose inputs form a finite set—each organ, cell type, graph edge and gene-set collection—on the FastAPI application in-process and saves the responses (1,339 files, 23.5 MB), and writes per-organ expression tables in compact form: fraction expressing as 8-bit and mean expression as 16-bit integers scaled to the organ maximum, compressed (7.6 MB and 16.9 MB). The signature lookup, gene and signature queries and the corresponding UMAP figures are re-implemented in the browser against these tables, and a test script compares them with the live service. The static build is deployed by a continuous-integration workflow to GitHub Pages; no server, account or login is involved.

3 Results

3.1 Overview of the resource

Cella is organised around five tools that share the harmonised cell-type registry (Figure 1, Table 1). The organ browser summarises each atlas (composition, UMAP, gene and signature views, transcription-factor and pathway activity, and ligand–receptor interactions). The marker atlas lists the gated discriminative markers of a cell type in every organ where it occurs and separates the universal core from organ-specific markers (Figure 2a). The fingerprint explorer shows signed weights side by side across organs, the relationship graph links related cell types across organs, and the signature lookup ranks cell types for a pasted gene list (Figure 2b).

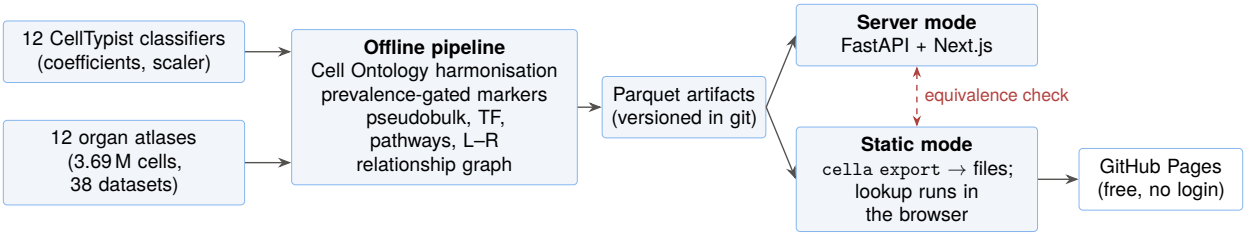


Figure 1. Architecture of Cella. An offline pipeline turns the CellTypist classifiers and atlases into versioned Parquet artifacts. The same interface is built either against a FastAPI service (server mode) or against pre-computed files with in-browser computation for free-text queries (static mode, used for the public site); a test compares the two modes' answers.

Table 1. The five tools of Cella. All tools share the Cell Ontology registry of 145 terms.

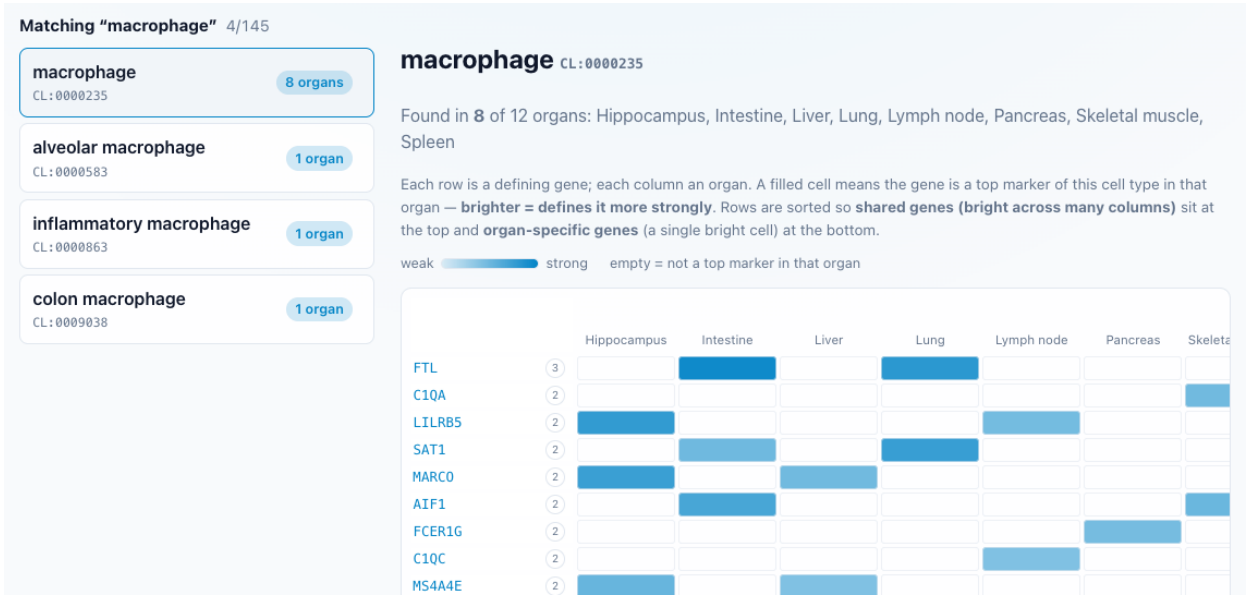
Tool	Question answered	Data used
Organ browser	What does this organ contain, and how is a gene or signature distributed across its cell types?	Atlas metadata, UMAP, pseudobulk, TF and pathway activities, ligand–receptor scores
Marker atlas	Which genes define this cell type in each organ, and which are shared?	Gated positive and negative coefficients
Fingerprint explorer	How does the weight of each defining gene change across organs?	Signed coefficients per organ
Relationship graph	Which cell types, in any organ, are defined similarly?	Cosine similarity of term-level weight vectors
Signature lookup	Which cell type does this gene list describe, and how confidently?	Fraction expressing (ranking); coefficients (evidence)

3.2 Raw classifier coefficients are dominated by rarely detected genes

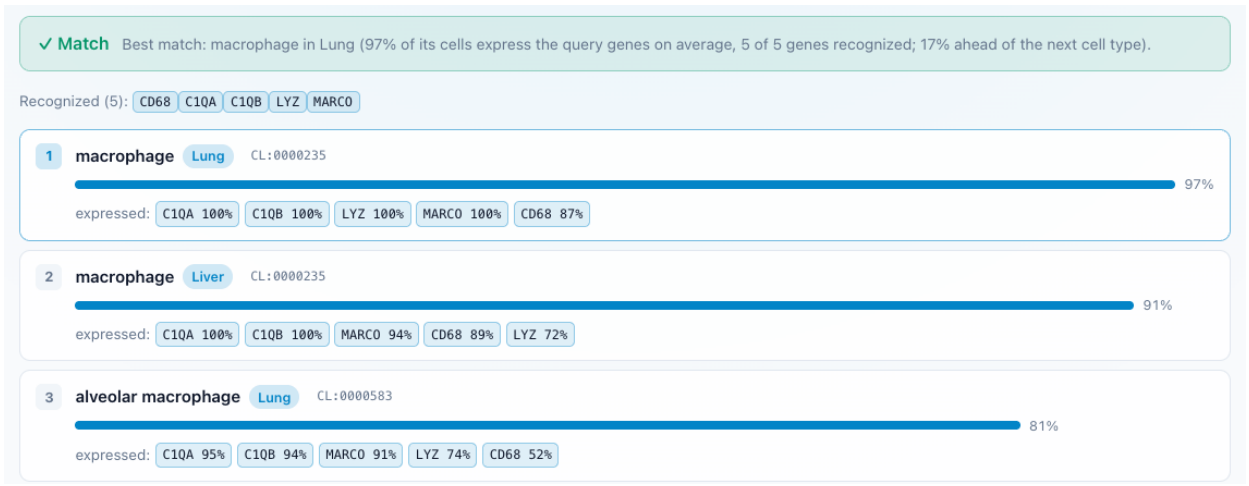
Ranking each label's genes by raw coefficient produced marker lists that were frequently implausible. Across the 399 organ-specific labels, 58% of the raw top-25 positive markers, and 27% of the top five, were genes expressed in fewer than 10% of the label's own cells; for 253 labels such genes made up more than half of the list (Figure 3a). The problem concentrated in rare labels (Spearman $\rho = -0.59$ between the contaminated fraction and the number of cells of the label in the atlas, $P = 2.4 \times 10^{-39}$; Figure 3b). This is consistent with the classifiers standardising each gene before fitting: a gene detected in a handful of cells has a small standard deviation, takes large standardised values in the few cells where it appears, and can carry a large coefficient that helps separate a small class without being a marker of it. Lung goblet cells, for example, ranked *FOXP3* among their top raw markers, and heart adipocytes ranked the natural-killer receptor *CD244* (Table 2). Prevalence gating removed these genes and brought canonical markers into the top 25, such as *MUC5AC* for goblet cells (raw rank 95, gated rank 10) and *PPARG* for adipocytes (raw rank 107, gated rank 10), while leaving well-represented labels such as alveolar macrophages unchanged.

3.3 Expression prevalence, not classifier weight, retrieves cell types from marker lists

The signature lookup was first implemented as the mean classifier weight of the query genes, on the reasoning that weights capture what defines a cell type. The benchmark did not support this choice (Figure 4a, Table 3). On the 231 held-out independent queries, mean weight placed a correct cell type first for 17% of queries (95% CI 12–23%) and among the top five for 32%. Gating the weights improved this to 22%, and per-gene standardisation of gated weights to 25%, while evaluating the classifier itself on a pseudo-cell expressing only the query genes performed worst (5%). Ranking by the mean fraction of cells expressing the query had the highest held-out accuracy, at 40% top-1 (95% CI 34–46%) and 73% top-5; it was correct where mean weight failed for 57 queries and the reverse for 5 (exact McNemar $P = 3.1 \times 10^{-12}$). Expression-based ranking beat mean weight in all 4 libraries, on queries overlapping the training data (66% versus 32%) and



(a)



(b)

Figure 2. The Cella interface (public static site). (**a**) Marker atlas for *macrophage*, found in eight organs: each row is a gated discriminative gene and each column an organ; shared genes are listed first. (**b**) Signature lookup for *CD68*, *C1QA*, *C1QB*, *LYZ* and *MARCO*: ranked hits with the fraction of cells expressing each gene; the alveolar macrophage hit is an ontology relative of the top hit and does not count against the confident call.

for short queries of three to ten genes (Figure 4b).

The advantage over the other expression-aware rules was not established. On held-out queries, mean fraction expressing was not significantly better than hypergeometric overlap with the gated markers (35% top-1; 34 versus 23 discordant queries, $P = 0.18$) or the hybrid score (32%; $P = 0.06$). Hypergeometric overlap had the higher top-1 accuracy in 3 of 4 libraries, including the development library (58% versus 50%, MRR 0.65 versus 0.63), and the pooled held-out lead of mean fraction expressing came from Tabula Sapiens (29% versus 16%, 116 queries; Azimuth 76% versus 78%, PanglaoDB 25% versus 32%). The development library therefore did not select mean fraction expressing by itself: no rule was best on every metric there, and mean fraction expressing was second on top-1 (tied with the hybrid score), top-5 and MRR (Table 3). We adopted it because it was consistently near the top, needs no marker cut-off and is directly interpretable; the held-out libraries confirm that it, like the other expression-aware rules, clearly outperforms classifier weights, but they do not single it out among those rules. A plausible explanation, which we did not test directly, is that logistic regression distributes weight across correlated genes, so canonical markers that co-vary with others receive modest individual coefficients, and that literature marker lists often include

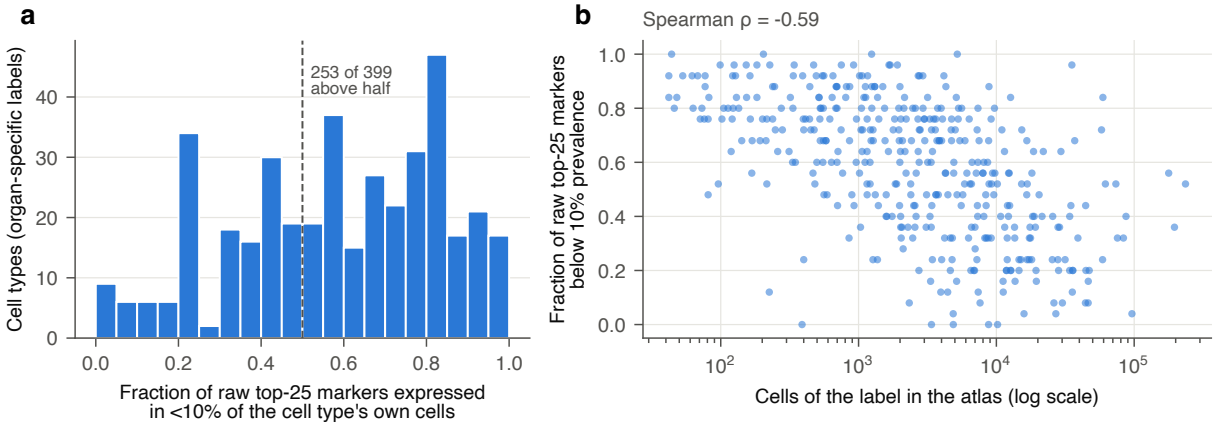


Figure 3. Raw classifier coefficients favour rarely detected genes. (**a**) For each of the 399 organ-specific labels, the fraction of its raw top-25 positive markers that are expressed in fewer than 10% of its own cells. (**b**) The same fraction against the number of cells of the label in the atlas.

Table 2. Top positive markers before and after prevalence gating. Genes expressed in fewer than 10% of the label's cells are shown in red.

Organ	Label	Raw coefficient ranking (top 6)	Gated ranking (top 6)
Lung	Goblet cells	*MSMB*, *CHST9*, *FOXP3*, *OVCH2*, *MIR202HG*, *LHX9*	*MSMB*, *CHST9*, *MUC16*, *NOS2*, *CYP2B6*, *CYP4F12*
Heart	Adipocytes	*ACACB*, *AGBL4*, *CD244*, *TNNI2*, *RP11-386G21.1*, *P2RY13*	*ACACB*, *AGBL4*, *GHR*, *ANGPTL8*, *GPAM*, *LIPE-AS1*
Blood	Age-associated B cells	*MS4A1*, *GALNTL6*, *C1QL3*, *PROB1*, *ARG1*, *SPOCK1*	*MS4A1*, *FGR*, *FCRL3*, *ENC1*, *RHOB*, *ZEB2*
Lung	Alveolar macrophages	*CTSD*, *APOC1*, *FTH1*, *FABP4*, *HLA-DRA*, *C1QA*	*CTSD*, *APOC1*, *FTH1*, *FABP4*, *HLA-DRA*, *C1QA*

broadly expressed genes that carry little discriminative weight. Cella therefore ranks lookups by expression and reports each gene's classifier weight alongside as evidence.

Because external marker lists are themselves usually derived from expression, this benchmark favours expression-based scoring by construction to some degree; it measures what users of a lookup tool need, namely retrieval from such lists, rather than the quality of the weights as descriptors. For the confidence call, the minimum score was fixed at 0.3 and the relative margin was examined on a grid from 0.05 to 0.25 (`supplementary_calibration.csv` in the repository). On the development library, precision rose and coverage fell monotonically with the margin, so no value was optimal; 0.10 was chosen as a compromise between the two. At these thresholds, 32% of held-out queries (73 of 231) were called confident, and their top hit was correct in 0.67 of cases (95% Wilson interval 0.56–0.77), against 0.29 (0.23–0.37) for the other 158 (development library: 0.68). Treating close ontology relatives as allies rather than competitors increased both the share of confident calls and their precision compared with requiring a lead over any other term. The calibration uses the benchmark's feature-set vocabulary and does not apply the minimum of two matched genes, so it approximates rather than reproduces the deployed call.

3.4 Gated coefficients make marker panels comparable to differential expression

On cells from datasets that neither method had seen, gated coefficient markers outperformed raw coefficient markers at every panel size and in both readouts (Figure 5; all $P_{\text{adj}} \leq 0.0076$). With ten markers per label and classifier-free assignment, gating raised macro-F1 by $+0.051$ (better in 44/48 splits, $P_{\text{adj}} = 1.5 \times 10^{-10}$), and by $+0.081$ with twenty markers (47/48, $P_{\text{adj}} = 7.3 \times 10^{-12}$). In the classifier-free readout, gated coefficients and Wilcoxon markers did not differ significantly at any panel size: the mean difference was -0.002, -0.009, $+0.004$ and $+0.016$ at $k = 3$, 5, 10 and 20 (unadjusted $P = 0.018$ at twenty, $P_{\text{adj}} = 0.2$; all other $P_{\text{adj}} \geq 0.81$). These are absence-of-difference results from 48 splits, not a formal equivalence test. Gated coefficients exceeded gated log fold change at ten and twenty markers ($P_{\text{adj}} = 0.0039$ and

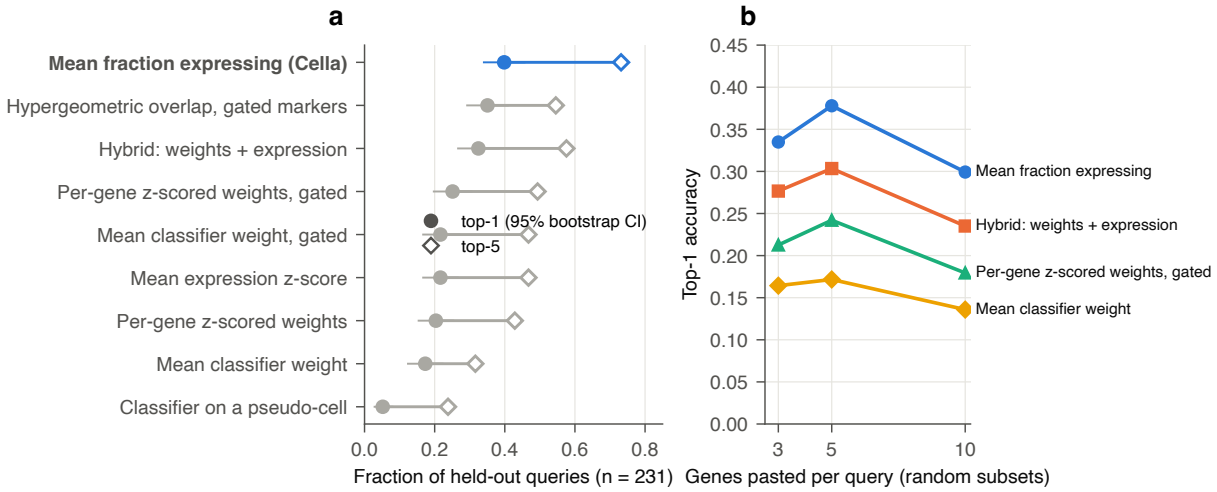


Figure 4. Benchmark of signature lookup against external marker sets. (**a**) Top-1 (filled, with 95% bootstrap interval) and top-5 (open) accuracy of nine scoring rules on 231 independent queries from Tabula Sapiens, Azimuth and PanglaoDB; CellMarker served as the development library. (**b**) Top-1 accuracy when each independent query is reduced to 3, 5 or 10 randomly chosen genes (five draws each).

Table 3. Lookup benchmark, independent queries. Development: CellMarker (150 queries). Held-out: Tabula Sapiens, Azimuth and PanglaoDB (231 queries). MRR, mean reciprocal rank.

	Development			Held-out		
Scoring rule	Top-1	Top-5	MRR	Top-1	Top-5	MRR
Mean fraction expressing (Cella)	0.50	0.77	0.63	0.40	0.73	0.54
Hypergeometric overlap, gated markers	0.58	0.73	0.65	0.35	0.55	0.46
Hybrid: weights + expression	0.50	0.77	0.62	0.32	0.58	0.45
Per-gene z-scored weights, gated	0.46	0.73	0.59	0.25	0.49	0.37
Mean expression z-score	0.48	0.79	0.62	0.22	0.47	0.34
Mean classifier weight, gated	0.44	0.74	0.58	0.22	0.47	0.34
Per-gene z-scored weights	0.37	0.62	0.49	0.20	0.43	0.31
Mean classifier weight	0.35	0.60	0.46	0.17	0.32	0.25
Classifier on a pseudo-cell	0.05	0.23	0.15	0.05	0.24	0.16

0.0025). When a classifier was trained on the panel, Wilcoxon markers were better with three markers per label (-0.034, better in 11/48 splits for gated coefficients, $P_{\text{adj}} = 1.3 \times 10^{-4}$); the difference at five markers (-0.021, unadjusted $P = 0.0093$) did not survive correction ($P_{\text{adj}} = 0.12$), and the methods converged at ten and twenty. The union panel of gated coefficients is larger at the same k (298 versus 218 genes at $k = 10$) because their per-label lists overlap less, which may favour them in this readout; the classifier-free readout uses exactly k genes per label and is the fairer comparison. Variation between held-out datasets was large relative to differences between methods (Figure 5), which is why we report paired tests. With 4 splits per organ the data do not support organ-level conclusions; Wilcoxon markers had the higher mean classifier-free macro-F1 at $k = 10$ in 5 of 12 organs.

3.5 Use case: organ invariance of discriminative identity

Comparing a cell type's weight vectors across organs asks whether the same genes distinguish it everywhere. Of 49 multi-organ terms, 30 are annotated at a specific level and 12 coarse terms could be compared on organs in which no subtype was split out; the other 7 were too coarse to compare (Figure 6). Calibration mattered: averaged over organ pairs, different cell types of two lymphoid organs had a mean cosine of 0.179 (6 pairs), against 0.101 for two non-lymphoid organs (27 pairs), so a single global null would have inflated the scores of cell types compared mostly across blood, bone marrow, spleen and lymph node. With

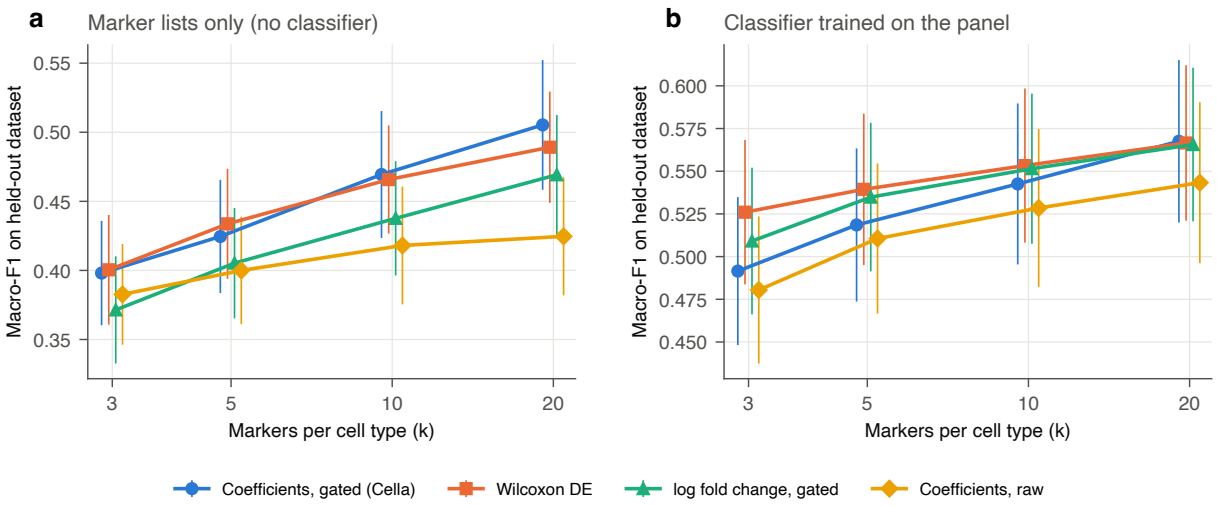


Figure 5. Marker panels evaluated on held-out datasets. Macro-F1 on cells of a held-out source dataset (48 splits, 12 organs; mean ± 1.96 standard errors across splits) for markers ranked by gated coefficients, Wilcoxon rank-sum, gated log fold change and raw coefficients. (**a**) Each cell is assigned to the label whose own k markers have the highest mean z-score. (**b**) A logistic regression is trained on the union of all labels' markers.

per-pair calibration, haematopoietic lineages were more organ-invariant than structural ones (median 0.177 versus 0.092; 22 versus 8 terms; two-sided Mann–Whitney $P = 0.0025$, compared with $P = 4.6 \times 10^{-5}$ before calibration). The haematopoietic group includes mast cells and conventional dendritic cells, which are tissue-resident; excluding them left the result unchanged (20 versus 8 terms, $P = 0.0039$). Restricted to pairs of non-lymphoid organs, the difference was in the same direction but not significant ($P = 0.059$, 14 versus 8 terms). Among the 30 specific terms, invariance was not associated with the number of cells (Spearman $\rho = -0.02$, $P = 0.90$) or, for the 28 terms with enough cells, with the reproducibility of the coefficients across disjoint training folds ($\rho = -0.15$, $P = 0.45$). This pattern is compatible with the view that tissue environment shapes the identity of resident cells, as shown for macrophage enhancer landscapes [19] and fibroblast states [20]. An alternative we cannot exclude is that immune populations were annotated more uniformly across the contributing datasets than structural ones. The analysis is small, rests on eight structural terms and one atlas collection, and we present it as an example of what the resource makes possible rather than as a definitive result.

3.6 A serverless deployment that matches the server

Because the public site has no backend, its answers had to be shown to match those of the service. On 126 lookups, comprising the six built-in examples and 120 benchmark queries, the in-browser implementation returned the same top hit as the service for 124; the remaining 2 were ties at the precision of the compact tables, where the two best scores differed by less than 0.004. The confidence call agreed for 126/126 queries, and scores differed by at most 0.0013. Gene-level values (five organ–gene queries) agreed to within 0.002 and signature scores (three queries) to within 0.001; graph slices had the same numbers of nodes and edges, and the sampled edge details, cell-type searches and per-gene cross-organ weights were identical. The site therefore costs nothing to host, has no cold start, and can be archived together with its code and data. One limitation is specific to static mode: classifier weights shown as evidence in lookup hits are available only for genes among the stored top markers, whereas the service reads the full coefficient matrix.

4 Discussion

Cella makes the classifiers behind a widely used annotation tool readable and queryable without code, and pairs them with the expression summaries needed to interpret them. Two findings shaped the resource and are

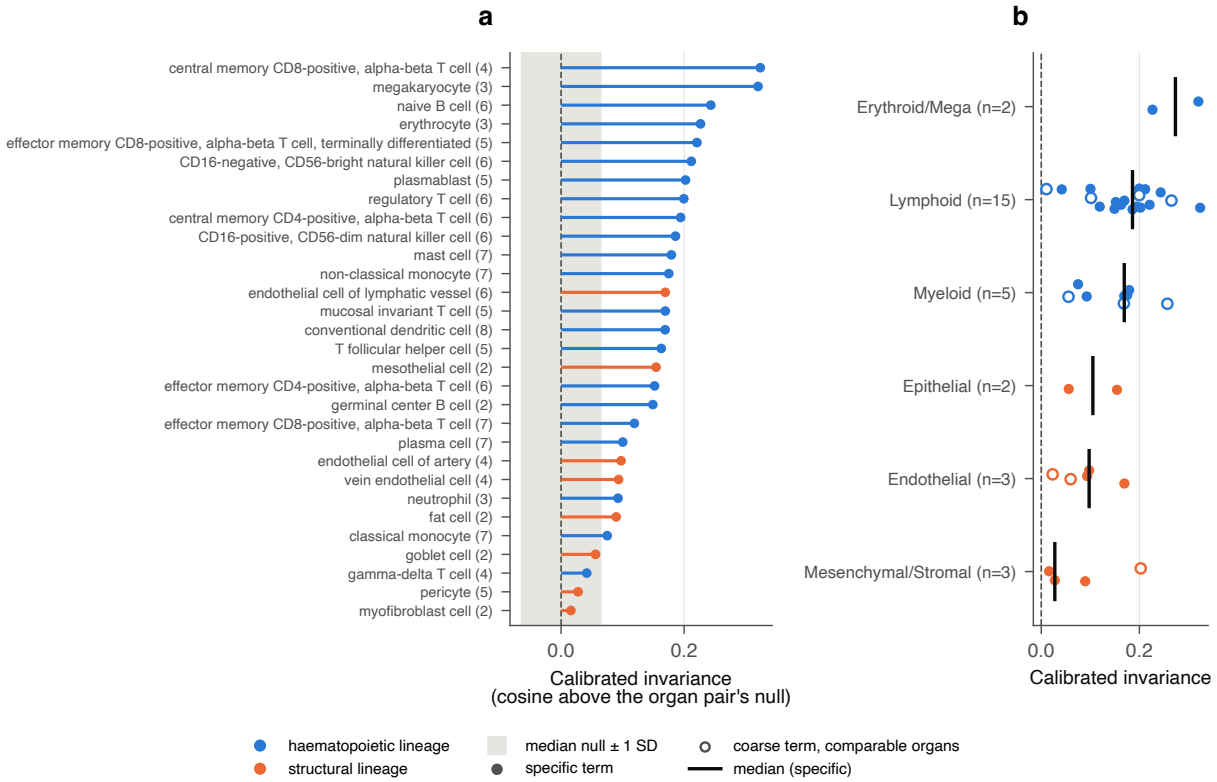


Figure 6. Organ invariance of discriminative identity. (**a**) Calibrated invariance (mean cross-organ cosine of weight vectors minus each organ pair's mismatched-cell-type null) for the 30 specifically annotated terms found in two or more organs; the number of organs is given in parentheses. Blue, haematopoietic lineages; orange, structural lineages; grey band, ± 1 s.d. of the median per-pair null (0.064; range across organ pairs 0.03–0.14). (**b**) The same scores by lineage, with the median of specific terms; open symbols, coarse terms scored on the organs in which no subtype was annotated.

relevant beyond it. First, coefficients of standardised logistic-regression models should not be ranked directly as markers: rarely detected genes acquire large weights for rare classes, and a simple prevalence gate turns the coefficients into marker lists that, in a classifier-free readout on unseen data, were not distinguishable from Wilcoxon differential expression, although Wilcoxon markers remained better for very small panels used to train a classifier. That the Wilcoxon test is hard to beat agrees with a systematic comparison of marker-selection methods [21]; the value of the gated coefficients lies less in outperforming it than in coming from the same model that annotates the data, so that the markers shown are the ones the classifier uses. Second, the classifier's discriminative weights are a poor basis for retrieving a cell type from a short marker list, where expression-aware scoring does markedly better; which expression-aware rule is best was not resolved by our benchmark. Discriminative weights and expression thus answer different questions, and the resource uses each where the benchmarks show it works: weights to describe and compare identity, expression to retrieve it.

Several limitations apply. The resource inherits the scope of the CellTypist organ atlases, including their tissue coverage, annotation granularity and gene vocabularies, and the relationship graph is a similarity graph over weight vectors rather than a reconstruction of the cross-dataset harmonisation of the original work [4]. The lookup benchmark depends on mapping free-text cell-type names to the Cell Ontology and on marker lists that are themselves largely expression-derived, and its accuracy figures should be read as relative rather than absolute. Marker-panel performance varied considerably between held-out datasets, each organ contributed only 4 splits, and each split was sampled once. The benchmarked lookup was restricted to the classifiers' gene vocabularies, whereas the deployed tool uses the full expression vocabulary, and the deployed confidence call adds a minimum number of matched genes, so the benchmark approximates the deployed tool rather than reproducing it. The invariance analysis rests on 30 specifically annotated terms, only 8 of them structural, and a single atlas collection; replication in an independent multi-organ atlas

profiled from the same donors, such as Tabula Sapiens [16], would be needed to draw biological conclusions.

Future work includes adding further atlases and model releases, recomputing the relationship graph with the original harmonisation approach, and offering programmatic access to the server mode. Because the deployment is static and versioned, each release of Cella can be archived and cited as a fixed snapshot.

Data and code availability

Cella is freely available without registration at https://mdjubayerhossain.com/cella/. Source code, the analysis and benchmark scripts, the processed artifacts and the scripts that generate every figure and number in this article are available at https://github.com/hossainlab/cella under the MIT licence. Input data are the public CellTypist organ atlases and models [5]; benchmark queries are public Enrichr libraries [15].

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Conflict of interest

None declared.

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